Sprint 1 Review & Sprint 2 Plan

Drawing Puzzle Game

Individual Sprint Review Presentation

- Sprint 1 Metrics
- Sprint 1 Retrospective
- Software Design Learning
- Sprint 2 Plan

Sprint 1 Metrics

Project Progress

Metric	Status
Lines of Code	~470
HW2 Feature Baseline	3 features
Current Approved Feature Scope	3 features
Completed Features	1 / 3 (33%)
HW2 Requirement Baseline	5 requirements
Current Approved Requirement Scope	5 requirements
Completed Requirements	3 / 5 (60%)

Scope Changes & Burndown

Scope Management

Scope change: Use a less sophisticated shape recognizer.

Reasons:

- A more advanced recognizer would add significant complexity.
- The added sophistication does not justify the implementation effort.
- The project timeline and deadline favor a simpler solution.

Approved scope remains unchanged:

- 3 features
- 5 requirements

Sprint 1 Retrospective

What Went Well

- The basic project structure went smoothly.
- The foundational framework is now in place.
- The project has a clearer path toward user-facing functionality.

What Went Wrong

- Certain features took longer than expected.
- Some features needed to be scaled down.
- The difficulty of some requirements was underestimated.

Main Lesson

Improvements & Development Progress

Cause of Challenges

Some requirements were more difficult to implement than expected within the available time.

Improvement Plan

- Evaluate technical complexity earlier.
- Prioritize essential functionality.
- Scale features to fit the project timeline.
- Focus on usability after establishing the core framework.

Course Weeks 5–8 Summary

Software Design Learning

SOLID Principles

Design principle learned: SOLID software design principles.

Why It Matters

- Encourages clear separation of concerns.
- Makes components easier to understand.
- Supports maintainable and extensible code.
- Reduces unnecessary coupling between responsibilities.

Application & Results

Applying SOLID principles helped guide the implementation toward clearer separation of responsibilities and a more organized project structure.

Sprint 2 Plan

MVP Goals

Primary goal: Create at least 3 playable puzzles with clear navigation.

MVP Objectives

- Create at least three playable puzzles.
- Implement clear navigation between game states and puzzles.
- Connect drawing interactions with in-game effects.
- Establish a functional puzzle creation framework.

Success Criteria

- Players can navigate the game clearly.

Sprint 2 Design & Features

Design Improvement Goal

Create prototype puzzles, test them with users, and improve the designs based on feedback.

Planned Features

- Multiple playable puzzles.
- Drawing interactions that trigger effects.
- Objects that respond to drawing.
- Clear navigation and improved user experience.

Planned Requirements

1. Build a puzzle creation framework

Sprint 2 Testing Strategy

Required Test Plan

Test Type	Implementation Plan
Unit Tests	Implement when the majority of the framework is finished.
Integration Tests	Implement when the majority of the framework is finished.
Regression Tests	Implement when the majority of the framework is finished.
Acceptance Tests	Implement when the majority of the framework is finished.

Testing Objectives

- **Unit testing:** Verify individual components and functions.
- **Integration testing:** Confirm that drawing, objects, and puzzle logic work together.

Sprint 2 Timeline: Weeks 9–10

MVP & Core Features

Planned Work

- Brainstorm puzzle concepts.
- Start creating drawing effects.
- Establish the initial playable puzzle structure.

Milestone

Establish the core gameplay foundation and begin implementing drawing-based interactions.

Sprint 2 Timeline: Weeks 11–13

Features, Testing & Design

Course Weeks 11–12

- Realize and implement the puzzles.
- Develop unit, integration, regression, and acceptance tests.
- Refine interactions and system design.

Course Week 13

- Conduct additional testing and acceptance checks.
- Improve existing systems.
- Prepare documentation.
- Create the initial GitHub Pages site.

Sprint 2 Timeline: Weeks 14–15

Fixes & Final Submission

Planned Work

- Resolve remaining bugs.
- Apply final gameplay and usability improvements.
- Complete project documentation.
- Prepare the final submission.

Final Milestone

Deliver a playable, tested, documented drawing puzzle game that meets the project's goals.

Sprint 1 Review & Sprint 2 Priorities

Sprint 1 Achievements

- Established the basic project framework.
- Completed 1 of 3 features (33%).
- Completed 3 of 5 requirements (60%).
- Learned to balance technical complexity with deadlines.
- Applied SOLID principles to guide software design.

Sprint 2 Priorities

1. Deliver at least three playable puzzles.
2. Connect drawing interactions to game effects.
3. Improve navigation and user experience.

References & Evidence

Project Repository

Drawing Puzzle Game — GitHub Commits

Evidence & Documentation

- Sprint 1 metrics and requirement burndown.
- Sprint 1 retrospective and software design learning.
- Sprint 2 feature, testing, and timeline plan.
- Weekly progress reports on Canvas.

Thank You!

Questions?